

FASSBERG, Germany

WMO: 102460

Lat: 52.9156N

Lon: 10.1886E

Elev: 88

StdP: 100.27

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-8.8	-13.6	1.2	-10.5	-10.8	1.5	-7.9	11.3	7.2	10.2	6.3	2.7	100	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.1	19.3	28.0	18.5	26.0	17.6	20.2	27.4	19.4	26.0	18.4	24.4	3.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	22.1	17.1	12.3	21.5	16.1	11.6	20.5	58.5	27.3	55.3	26.2	52.3	24.1	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.1	7.2	DB	-15.1	34.1	4.2	2.2	-18.1	35.7	-20.6	37.0	-23.0	38.2	-26.1	39.8	(4)
(5)			WB	-15.3	22.4	4.2	1.2	-18.3	23.2	-20.7	24.0	-23.0	24.6	-26.1	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	0.5	1.7	4.0	8.4	12.6	15.7	17.9	17.6	13.6	9.4	4.8	1.7
	DBStd	7.27	4.92	4.55	3.93	3.93	3.80	3.33	3.20	3.07	3.17	3.69	3.83	4.70
	HDD10.0	1286	294	234	187	76	16	1	0	0	5	56	159	257
	HDD18.3	3479	553	467	443	297	183	94	48	51	146	278	406	514
	CDD10.0	933	0	1	3	28	96	170	245	237	113	37	3	1
	CDD18.3	85	0	0	0	0	4	14	34	29	3	0	0	0
	CDH23.3	1126	0	0	0	7	87	208	412	349	60	2	0	0
	CDH26.7	333	0	0	0	1	17	59	132	111	13	0	0	0
Wind	WSAvg	3.3	3.6	3.7	3.7	3.3	3.3	3.2	3.1	2.9	2.9	3.2	3.2	3.5
Precipitation	PrecAvg	777	76	56	56	45	58	68	76	73	62	61	66	76
	PrecMax	1041	158	123	133	96	163	133	183	150	197	185	128	171
	PrecMin	554	5	16	8	9	17	27	21	24	16	20	3	26
	PrecStd	130	40	27	34	21	32	28	39	33	40	32	32	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	14.2	18.6	24.1	28.9	32.0	33.0	32.9	28.9	22.0	15.0	12.8
		MCWB	9.6	8.1	11.1	14.5	17.9	20.7	19.5	19.9	18.2	16.1	12.3	10.6
	2%	DB	9.9	11.6	15.1	21.1	25.8	28.2	30.1	29.7	25.0	18.8	13.1	11.0
		MCWB	8.1	8.3	9.4	12.9	16.6	18.6	19.5	18.9	17.2	14.7	11.1	9.5
	5%	DB	8.3	9.8	12.3	19.0	23.1	25.7	27.8	27.2	22.0	16.7	11.6	9.2
		MCWB	7.0	7.6	8.3	11.6	14.9	17.2	18.6	18.4	15.8	13.7	9.6	8.0
	10%	DB	6.9	8.2	10.3	16.6	20.9	23.2	25.3	24.8	19.9	15.0	10.1	7.9
		MCWB	5.7	6.2	7.5	10.9	14.0	16.0	17.5	17.5	15.0	12.8	8.7	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.1	11.9	15.5	19.2	21.3	21.6	21.4	19.3	16.7	13.3	11.1
		MCDB	11.0	12.2	16.7	21.9	26.8	28.6	28.5	28.7	26.0	20.3	14.6	12.5
	2%	WB	8.4	8.8	10.4	13.9	17.4	19.6	20.2	20.1	17.7	15.2	11.4	9.6
		MCDB	9.6	10.9	13.5	19.1	24.1	26.6	27.5	26.7	22.9	17.9	12.9	10.6
	5%	WB	7.0	7.4	9.1	12.6	15.9	18.1	19.3	19.1	16.6	14.1	10.0	8.2
		MCDB	8.2	9.4	11.9	17.7	21.6	23.8	26.2	25.3	21.0	16.3	11.1	9.3
	10%	WB	5.6	6.0	7.7	11.4	14.5	16.9	18.3	18.2	15.6	12.9	8.7	6.7
		MCDB	6.6	8.0	10.1	15.9	19.5	22.1	24.2	23.5	19.0	14.8	9.9	7.5
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.6	8.6	11.8	11.7	11.6	11.0	11.5	10.6	8.2	5.6	4.8
		MCDBR	6.0	8.3	13.1	16.0	16.0	16.0	15.5	15.8	14.6	10.6	6.5	5.4
		MCWBR	5.1	5.7	8.4	8.8	8.3	7.9	6.7	7.0	8.1	7.1	5.0	4.8
	5% WB	MCDBR	5.4	7.0	11.1	13.9	14.1	14.0	13.5	13.6	12.3	9.8	5.9	5.4
		MCWBR	4.8	5.3	7.6	8.0	7.7	7.5	6.3	6.9	7.2	6.8	5.0	5.0
Clear-Sky Solar Irradiance	taub	0.315	0.335	0.367	0.410	0.408	0.406	0.423	0.416	0.385	0.353	0.332	0.309	
	taud	2.369	2.363	2.295	2.218	2.265	2.292	2.254	2.289	2.364	2.437	2.411	2.360	
	Ebn at Noon	667	768	812	820	842	847	822	806	787	739	641	601	
	Edh at Noon	59	79	103	126	126	124	127	116	96	73	57	50	
All-Sky Solar Radiation	RadAvg	0.62	1.31	2.42	3.99	4.94	5.26	4.97	4.17	3.05	1.71	0.75	0.47	
	RadStd	0.07	0.18	0.24	0.49	0.57	0.41	0.69	0.33	0.35	0.18	0.08	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-153	N/A	N/A

Nomenclature: See separate page